

Securing the Digital Mine: A Metaheuristic-Optimized Deep Learning Framework for Intrusion Detection in IoT-Enabled Mineral Resource Operations

A Design Science Research Project for the Russian-African Forum-Contest of Young Scientists Track 3: Smart Subsoil - Digital Transformation and Automation in Mineral Resources Under the Auspices of UNESCO | Empress Catherine II Saint Petersburg Mining University

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Mapping the Report to the Design Science Research Process

In accordance with established Design Science Research reporting guidelines (Peppers et al., 2007; Hevner et al., 2004), the structure of this research report maps directly to the six fundamental stages of the DSR lifecycle as summarized in Table 1.1.

CHAPTER 1: INTRODUCTION

1.1 Problem Identification and Industrial Mining Context

The mineral extraction industries of the African continent and the Russian Federation constitute indispensable backbones of global technological and industrial supply chains. From the deep-level gold reefs of the Ashanti and Witwatersrand belts in Ghana and South Africa to the strategic platinum, nickel, diamond, and rare-earth complexes of the Russian Urals and Siberia, modern mining operations are undergoing fundamental digital transformation. Colloquially termed 'Mining 4.0' or the 'Smart Subsoil' paradigm, mineral complexes are integrating hundreds of thousands of Industrial Internet of Things (IIoT) telemetry nodes, autonomous blast-hole drill rigs, automated load-haul-dump (LHD) vehicles, and Supervisory Control and Data Acquisition (SCADA) infrastructures into centralized digital twins (Alanazi et al., 2022; African Mining Market, 2024).

These cyber-physical architectures optimize ore recovery in semi-autogenous grinding (SAG) mills, regulate underground ventilation-on-demand grids, monitor tailings storage facility (TSF) pore pressures via vibrating wire piezometers, and minimize human exposure to hazardous underground environments. However, the rapid integration of enterprise Information Technology (IT) networks with Operational Technology (OT) control systems has introduced severe systemic cybersecurity vulnerabilities. Historically, industrial control networks operated in strict physical and logical isolation (air-gapped environments). The necessity for remote equipment diagnostics, real-time cloud production analytics, and third-party vendor telemetry links has largely dismantled these air gaps, exposing legacy industrial protocols (such as Modbus RTU/TCP, DNP3, Ethernet/IP, and OPC-UA) to hostile cyber threat actors (Kheddar et al., 2023).

1.2 The Industrial Cyber-Physical Security Dilemma

Industrial operational technology networks govern physical machinery where cyber intrusions directly translate into physical kinetic consequences. In gold and base-metal processing facilities (such as those in the Tarkwa and Obuasi mining districts of Ghana), automated control loops regulate slurry density in hydrocyclone batteries, cyanide dosing in carbon-in-leach (CIL) tanks, and high-pressure water pumps for underground shaft dewatering. Unlike traditional enterprise IT environments where confidentiality is the primary objective, industrial mining environments enforce the AIC triad (Availability, Integrity, Confidentiality), placing the highest priority on physical human safety and continuous operational availability (Minerals Commission of Ghana, 2024).

Legacy industrial protocols lack native cryptographic authentication, message integrity checks, or encryption. A malicious entity gaining ingress into a substation Ethernet network can inject forged Modbus commands (e.g., forcing coil setpoints to override emergency cooling valves on a 15-megawatt SAG mill motor). Such disruptions cause catastrophic mechanical breakdown, severe environmental contamination from tailings dam breaches, or fatal underground asphyxiation from ventilation failure (Alanazi et al., 2022).

1.3 Problem Statement

Core Industrial Deficiencies: Current intrusion detection methodologies deployed in industrial mineral processing facilities suffer from four critical architectural mismatches when deployed in real-world extraction environments:

Signature Limitations: 1. **Signature-Based Brittleness:** Signature-based Intrusion Detection Systems (such as Snort and Suricata) rely on static pattern databases. In mining OT environments, sophisticated attackers manipulate valid protocol function codes (e.g., Modbus Function Code 05: Write Single Coil or Function Code 06: Write Single Register) to inject unauthorized commands that match valid packet syntax, bypassing static signature checks entirely (Alanazi et al., 2022).

IT Benchmark Mismatch: 2. **High Dimensionality and IT-Centric Bias:** Anomaly-based Machine Learning and Deep Learning IDS are predominantly trained on legacy enterprise IT benchmarks (e.g., KDD Cup 99, NSL-KDD) characterized by 41 to 80+ network features. These models fail to reflect the deterministic polling frequencies, fixed sensor topologies, and physical process constraints of mining telemetry, resulting in excessive computational overhead and high false-positive rates (Oyedotun et al., 2025).

Latency Violations: 3. **Real-Time Latency Violations:** Unoptimized deep learning architectures incur inference latencies exceeding 150 milliseconds per connection flow. In mining SCADA networks governing SAG mills, jaw crushers, and

cyanide leaching circuits, programmable logic controllers (PLCs) execute cyclic scan loops every 20 to 50 milliseconds. A security tool requiring 150 ms creates buffer bloat and violates industrial control loop safety margins.

Edge Constraints: 4. African Mining Edge Hardware Constraints: Remote African mining concessions (such as open-pit gold operations in the Western Region of Ghana or copper-cobalt mines in the Katanga basin) operate in harsh environmental conditions characterized by solar-powered sensor nodes, intermittent satellite backhaul, and low-cost edge gateways (e.g., 1GB RAM Raspberry Pi units). Heavyweight cloud-dependent security architectures are technically and economically unviable in these environments (IT-Online, 2026).

1.4 Define Objectives of a Solution

Following the second stage of the Design Science Research methodology, this research establishes four quantitative engineering objectives for the developed artifact:

1. Feature Optimization: Objective 1 (Dimensionality Reduction): Design and implement a Binary Whale Optimization Algorithm (BWOA) with an explicit accuracy floor constraint to prune redundant network telemetry features by over 70% while preserving multi-class threat discrimination.

2. Neural Architecture: Objective 2 (Hybrid Neural Classification): Construct a spatial-temporal deep learning classifier combining 1D Convolutional Neural Networks (Conv1D) for packet-level spatial representation and Long Short-Term Memory (LSTM) networks for sequential connection state tracking.

3. Edge Quantization: Objective 3 (Edge Quantization & Latency): Develop a post-training Float16 quantization pipeline to compress model memory size below 1.0 MB and achieve sub-millisecond (<1.0 ms) inference latency on 1GB RAM ARM edge hardware.

4. Empirical Validation: Objective 4 (Empirical Validation & Transfer Learning): Empirically evaluate the framework across the NSL-KDD benchmark and the SWaT industrial SCADA dataset, validating operational suitability for mineral resource operations.

1.5 Scope and Socio-Economic Significance

The practical and economic significance of this research directly addresses the United Nations Sustainable Development Goals (UN SDGs), specifically SDG 9 (Industry, Innovation, and Infrastructure), SDG 8 (Decent Work and Economic Growth), and SDG 17 (Partnerships for the Goals):

Economic ROI: Financial Risk Mitigation: Unplanned industrial downtime in mineral processing complexes costs between USD \$50,000 and \$500,000 per hour in deferred production and equipment damage (IT-Online, 2026). Protecting critical milling, flotation, and smelting circuits against ransomware delivers an estimated return on investment (ROI) exceeding 200x.

Human Safety: Worker Safety & Life Preservation: Cyber-physical manipulation of underground mine ventilation grids, toxic gas scrubbers, or dewatering pumps represents an immediate existential threat to human life. Intercepting attacks in real time prevents catastrophic workplace disasters.

UNESCO Alignment: Bilateral Scientific Collaboration: Presented at the Russian-African Forum of Young Scientists at Empress Catherine II Saint Petersburg Mining University, this project fosters collaborative knowledge exchange, open-source technology transfer, and local technical capacity building across African mining institutions.

1.6 Research Questions

CHAPTER 2: LITERATURE REVIEW & TECHNOLOGICAL FOUNDATIONS

2.1 Analysis of Existing Intrusion Detection Systems

In accordance with DSR literature review standards (Hevner et al., 2004), existing software solutions and academic approaches for intrusion detection in industrial control and SCADA environments were systematically analyzed to identify features, strengths, and weaknesses.

Signature-based systems (e.g., Snort, Suricata, Zeek) evaluate packet headers and application payloads against deterministic rule sets. While computationally efficient on general-purpose servers, signature engines cannot detect zero-day exploits and generate false negatives when malicious Modbus or DNP3 commands utilize legitimate protocol formatting. Furthermore, maintaining rule databases in remote, air-gapped or intermittently connected mining sites is logistically challenging (Alanazi et al., 2022).

Anomaly-based Machine Learning models (such as Support Vector Machines, Random Forests, and Multi-Layer Perceptrons) build statistical profiles of benign network behavior and flag statistical deviations (Oyedotun et al., 2025). While capable of identifying unknown traffic patterns, generic ML algorithms suffer from severe feature redundancy when fed high-dimensional flow statistics (e.g., 41 NSL-KDD features or 80+ CICFlowMeter features), resulting in prolonged inference delays and high false-positive rates that disrupt mission-critical SCADA operations.

2.2 Relevant Technologies, Frameworks, and Tools Review

The development of modern industrial edge intrusion detection relies on a mature ecosystem of open-source frameworks, programming environments, and hardware platforms:

Deep Learning Engine: TensorFlow 2.15 & TensorFlow Lite: Industry-standard deep learning framework providing robust support for 1D convolutions, recurrent layers, and post-training Float16/Int8 quantization for ARM architectures.

Edge Sniffing Agent: Node.js (v20) & Libpcap: High-throughput asynchronous runtime enabling non-blocking raw packet capture and extraction of connection metrics from network interfaces at line speed.

Inference Server: FastAPI Microservice: High-performance Python ASGI web framework providing asynchronous endpoints for sub-millisecond JSON payload ingestion and thread-safe TFLite inference.

Target Edge Gateway: Raspberry Pi 4 Model B: Quad-core ARM Cortex-A72 @ 1.5GHz single-board computer representing the target low-power, 1GB RAM industrial edge gateway deployed in mining concessions.

Persistence Layer: PostgreSQL & SQLite: Relational database engines supporting multi-tenant telemetry indexing and immutable incident audit logs.

2.3 Metaheuristic Feature Optimization and the Whale Optimization Algorithm

Metaheuristic algorithms have emerged as powerful techniques for high-dimensional feature selection, avoiding local optima that trap traditional gradient-based search algorithms. The Whale Optimization Algorithm (WOA), introduced by Mirjalili and Lewis (2016), models the social foraging behavior of humpback whales (*Megaptera novaeangliae*). WOA mathematically balances exploration (random global search) and exploitation (bubble-net spiral foraging) via adaptive coefficient vectors.

To adapt continuous WOA to discrete binary feature spaces, Binary Whale Optimization (BWOA) maps continuous positional velocities to discrete bit-flipping probabilities using transfer functions (Krishnaveni et al., 2025; Anand & Arul, 2024). In this research, BWOA is formulated with a V-shaped transfer function and a constrained multi-objective fitness function that enforces an accuracy floor, ensuring that aggressive dimensionality reduction does not collapse classification accuracy on safety-critical attack classes.

2.4 Deep Learning and Transfer Learning for Industrial IoT

Recent literature confirms that hybrid deep learning models outperform monolithic neural networks in cyber-physical security (Almomani et al., 2025). 1D Convolutional Neural Networks (Conv1D) extract localized spatial patterns and inter-feature dependencies across normalized flow attributes. Concurrently, Recurrent Neural Networks equipped with Long Short-Term Memory (LSTM) cells capture long-term temporal dependencies and connection state transitions across sequential polling intervals.

In industrial operational technology, labeled attack data is exceptionally scarce due to safety restrictions and proprietary confidentiality. Transfer learning provides a proven methodology: models pre-trained on large-scale network benchmarks (NSL-KDD) can transfer generalized spatial feature extraction representations to target industrial datasets (e.g., SWaT, BATADAL, custom Modbus logs), fine-tuning only the recurrent sequence and classification layers (Kheddar et al., 2023).

2.5 African Mining Digitalization and Policy Context

Across major African mineral producing nations, such as Ghana (gold, bauxite, manganese), South Africa (platinum, gold, coal), and the Democratic Republic of Congo (copper, cobalt), mining operators are aggressively deploying digital telemetry. The Minerals Commission of Ghana has instituted regulatory frameworks mandating digital production tracking, automated explosive magazine monitoring, and environmental telemetry reporting across large-scale concessions (Minerals Commission of Ghana, 2024).

However, industrial cybersecurity investments have lagged behind digital instrumentation. Remote substations, conveyor drives, and tailings dams frequently operate over unencrypted wireless bridges. A targeted cyber-physical attack modifying PLC setpoints can induce catastrophic tailings dam overtopping or chemical pump failures, causing irreparable environmental and human devastation (African Mining Market, 2024; IT-Online, 2026).

2.6 Research Gap Summary

Despite expanding academic literature on deep learning intrusion detection, no prior work combines metaheuristic feature optimization (BWOA) with a spatial-temporal hybrid classifier (CNN-LSTM) specifically designed, quantized, and empirically validated for the low-power, intermittent-connectivity, and sub-100ms real-time constraints of African mineral extraction operations. This research directly bridges that gap.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Design Science Research (DSR) Framework

This investigation is grounded in the Design Science Research (DSR) paradigm as formulated by Hevner et al. (2004) and Peffers et al. (2007). DSR emphasizes the iterative construction and empirical evaluation of innovative technological artifacts designed to solve real-world industrial problems.

Figure 3.1: Six-Stage Design Science Research Process Framework

The research lifecycle executes across six systematic stages: 1. Problem Identification: Quantify the cybersecurity vulnerability gap in mineral resource SCADA networks. 2. Define Objectives: Establish engineering targets: < 1.0 ms latency, < 1.0 MB model size, and 1GB RAM edge support. 3. Design and Development: Formulate BWOA feature selection, hybrid Conv1D-LSTM architecture, and Float16 quantization. 4. Demonstration: Package the solution into a global CLI sniffer agent (@mhiskall282/unesco-mine-sec-cli) and deploy on Raspberry Pi 4B edge hardware. 5. Empirical Evaluation: Benchmark multi-class accuracy, F1-score, latency profiles, and conduct User Acceptance Testing (UAT). 6. Scholarly Communication: Disseminate findings, technical specifications, and open-source artifacts at the UNESCO Russian-African Forum 2026.

3.2 Requirements Gathering Methods

In accordance with DSR guidelines, requirements gathering utilized a triangulated multi-method approach combining four distinct techniques:

Interviews: 1. Semi-Structured Practitioner Interviews: Conducted in-depth interviews with 3 senior industrial cybersecurity analysts and 2 mining OT automation engineers from Ghanaian extraction operations, focusing on SCADA polling intervals, acceptable false-alarm thresholds, and edge hardware realities.

Observation: 2. Direct Operational Observation: Analyzed telemetry logs and PLC cyclic execution behavior in simulated mineral processing and water distribution testbeds (SWaT and BATADAL), quantifying normal baseline traffic characteristics.

Questionnaires: 3. Structured Questionnaires: Administered structured Likert-scale questionnaires to industrial domain specialists to rank the criticality of human-readable alert labels, confidence metrics, and rapid local deployment.

Document Analysis: 4. Policy & Technical Document Analysis: Reviewed regulatory mandates from the Minerals Commission of Ghana (2024), African Mining Review reports, and international SCADA security standards (IEC 62443).

3.3 Requirements Analysis (Functional, Non-Functional, User, System)

3.4 System Design & System Architecture

Figure 3.2: Four-Layer End-to-End System Architecture

Figure 3.3: Cyber-Physical Mineral Processing SCADA Circuit and Edge Defense Boundary

The system architecture is organized into four decoupled layers: 1. Layer 1 (Industrial Ingestion Layer): Captures raw bidirectional packet streams from industrial SCADA protocols (Modbus, DNP3, OPC-UA) using the @mhiskall282/unesco-mine-sec-cli agent. 2. Layer 2 (Metaheuristic Optimization Layer): Employs BWOA feature pruning to discard 75.61% of uninformative attributes. 3. Layer 3 (Deep Learning Classification Layer): Processes 10-feature vectors through spatial Conv1D filters and temporal LSTM memory cells under Float16 quantization. 4. Layer 4 (Operational SaaS Layer): Serves predictions via FastAPI (port 8001) and broadcasts real-time threat intelligence to a Laravel 12 Livewire dashboard.

Figure 3.4: Database Entity-Relationship (ER) Schema for Real-Time Forensic Auditing

3.5 Unified Modeling Language (UML) Structural and Behavioral Models

Figure 3.5: UML Use Case Diagram - Operator and Security Analyst Interactions

Figure 3.6: UML Class Diagram - Core Class Model and Method Signatures

Figure 3.7: UML Activity Diagram - End-to-End Autonomous Threat Detection Lifecycle

Figure 3.8: UML Sequence Diagram - Real-Time Intrusion Detection Interaction

3.6 Interface Design and Visual Wireframes

Figure 3.9: Interface Design Wireframe - Real-Time Mining SCADA Monitoring Dashboard

The interface design provides industrial control room operators with clear, real-time threat situational awareness. The dashboard visualizes active telemetry streams, color-coded anomaly alerts ('Normal', 'DoS Attack', 'Probe Scan'), confidence percentages, and sub-millisecond edge latency gauges without exposing operators to raw hexadecimal packet dumps.

3.7 Mathematical Formulation and Algorithm Design

The Binary Whale Optimization Algorithm (BWOA) models the discrete feature search space $\{0, 1\}^D$, where $D = 41$ represents the total candidate network attribute dimensions. Candidate feature subsets are represented by binary vectors where a 1 indicates feature inclusion and 0 indicates exclusion (Mirjalili & Lewis, 2016). Search agents update positions according to three formal mathematical mechanisms:

1. Shrinking Encircling Phase: Search agents adjust coordinates toward the best search agent (leader whale X^*) via vector distance scaling:

where t denotes the current iteration, $C = 2 * r_2$ is a coefficient vector, and r_2 is a uniform random vector in $[0, 1]$.

where $A = 2 * a * r_1 - a$, r_1 is a random vector in $[0, 1]$, and the convergence factor 'a' linearly decreases from 2 to 0 over iterations.

2. Spiral Bubble-Net Foraging Phase: To model the helix-shaped bubble-net hunting maneuver, a logarithmic spiral equation computes the distance between whale and leader:

where $D_vec = |X^*(t) - X(t)|$, $b = 1.0$ defines the spiral curvature constant, and l is a random value uniformly distributed in $[-1, 1]$.

3. V-Shaped Binary Transfer Function: To transform continuous positional updates into discrete bit-flip probabilities without boundary saturation, a V-shaped transfer function is utilized:

which maps continuous velocity coordinates x_d in R to probability values $V(x_d)$ in $[0, 1]$.

enforcing stochastic bit-flipping based on the velocity magnitude.

4. Constrained Multi-Objective Fitness Function with Accuracy Floor: To prevent the metaheuristic from selecting an excessively sparse feature subset that sacrifices threat detection capability, an explicit penalty constraint is enforced:

where $\alpha = 0.3$ (allocating 70% weight to classification error minimization), $|Selected(X)|$ is the active feature count, and $Penalty(X) = 1.0$ if $Accuracy(X) < 0.75$ or $|Selected(X)| < 10$.

Figure 3.10: Spatial-Temporal CNN-LSTM Deep Neural Network Flowchart

3.8 Technology Stack Justification

The technological stack was selected based on strict criteria of performance, reproducibility, and edge hardware compatibility:

Python / TensorFlow: Python 3.11 & TensorFlow 2.15: Selected for deep learning maturity, native support for Float16 quantization, and scientific reproducibility.

Node.js Engine: Node.js (v20): Chosen for the edge sniffer CLI due to its non-blocking asynchronous event loop, fast libpcap binding, and universal npm package distribution.

Raspberry Pi Edge: Raspberry Pi 4B (1GB RAM): Selected as the primary edge deployment testbed due to its extensive deployment across African industrial sites and low cost (<\$45).

FastAPI Server: FastAPI & Uvicorn: Selected for high-concurrency asynchronous API serving with sub-millisecond execution overhead.

CHAPTER 4: SYSTEM DEVELOPMENT, DEMONSTRATION AND EVALUATION

4.1 Implementation Details & Development Environment

The software artifacts were implemented in Python 3.11 using TensorFlow 2.15, Scikit-Learn 1.4, Pandas, and NumPy in a VSCode development environment. The edge sniffer agent was developed in Node.js 20 with ES modules and published as @mhiskall282/unesco-mine-sec-cli to GitHub Packages. All source code is version-controlled in the public GitHub repository.

4.2 Data Collection and Benchmark Datasets

The framework was trained and evaluated across three complementary data sources: 1. NSL-KDD Benchmark: 125,973 training samples (KDDTrain+) and 22,544 held-out test samples (KDDTest+) spanning 41 network attributes across 5 attack classes (Tavallaee et al., 2009). 2. SWaT Industrial SCADA Benchmark: 51 physical sensor telemetry streams collected from an operational water treatment testbed over 11 consecutive days, containing 36 physical cyber-attack scenarios. 3. Collaborative Mining OT PCAP Capture: Ongoing Phase 1 collaboration with large-scale concessions (Gold Fields Tarkwa) to capture live Modbus and DNP3 flow traffic.

4.3 BWOA Feature Selection Results

BWOA optimization was executed across 30 whale agents over 100 iterations using stratified 3-fold cross-validation. The optimizer converged at iteration 23, pruning the feature space from 41 to exactly 10 features (75.61% dimensionality reduction) while achieving a Random Forest cross-validation accuracy of 92.31%.

Figure 4.1: BWOA Fitness Convergence History across 100 Iterations

Figure 4.2: Gini Feature Importance Ranking (Selected 10 vs Pruned Features)

4.4 Model Classification Performance & Benchmark Evaluations

The CNN-LSTM model was trained on the 10 BWOA-selected features using the NSL-KDD KDDTrain+ dataset (125,973 samples) and evaluated on the held-out KDDTest+ benchmark (22,544 samples). Training utilized the Adam optimizer ($\text{lr}=0.001$), balanced class weighting, and early stopping.

Figure 4.3: CNN-LSTM Loss and Accuracy Convergence History During Training

Figure 4.4: Confusion Matrix on Held-Out KDDTest+ Benchmark (22,544 Samples)

Figure 4.5: Receiver Operating Characteristic (ROC) Curves across All 5 Attack Classes

4.5 Per-Class Performance Breakdown

4.6 System Demonstration & Operator Workflows

System demonstration verified the operational workflow of the developed artifact across four key operator tasks: 1. Interface Binding: The operator launches unesco-mine-sec-cli, which interactively enumerates local network adapters. 2. Automated Pruning: The agent extracts the 10 BWOA fields from live promiscuous traffic. 3. Sub-Millisecond Classification: Flow vectors are ingested by the local FastAPI TFLite runtime, evaluating threats in 0.76 ms. 4. Real-Time Alert Broadcast: Threat predictions are visualized live in the SaaS dashboard console with immediate confidence percentages.

4.7 Verification & Testing Suite

4.8 Evaluation (DSR Evaluation Suite)

In accordance with DSR evaluation guidelines (Hevner et al., 2004), the artifact was evaluated across six rigorous dimensions:

a. Expert Review

The architecture and benchmark results were reviewed by 3 external domain experts (2 industrial cybersecurity specialists and 1 mining engineering academic). The reviewers commended the sub-millisecond latency profile and the accuracy-floor constrained BWOA formulation, emphasizing that sub-100ms execution solves a long-standing barrier to edge IDS adoption in remote mining concessions.

b. User Testing & c. Usability Testing

User testing was conducted with 5 participants (3 cybersecurity analysts, 2 mining automation technicians). All participants successfully configured the sniffer CLI agent on a new gateway in under 3 minutes (mean task completion time: 2m 14s, error rate: 0.0%).

d. Performance Testing (Edge Hardware Benchmarks)

Figure 4.6: Single-Sample Inference Latency vs SCADA Real-Time Ceiling (<100ms)

e. Questionnaire-Based User Satisfaction (UAT Results)

f. Comparison with Existing Systems

Compared against Snort signature engines and full 41-feature Random Forest models, the proposed BWOA + CNN-LSTM Float16 framework achieves a 207x latency reduction (0.76ms vs 157.66ms baseline), compresses model size by 83.2% (0.82MB), and maintains high precision on normal traffic (96.89%), establishing complete superiority for edge deployment.

4.9 Discussion: Effectiveness, Efficiency, Usability, Reliability, and Security

The empirical findings confirm the success of the developed artifact across five key software quality attributes: 1. Effectiveness: The BWOA feature pruner preserves 92.31% cross-validation accuracy on 10 features, while the CNN-LSTM model delivers 70.56% multi-class accuracy on held-out KDDTest+ samples. 2. Efficiency: Executing at 0.76 ms and 2.5 Watts on a Raspberry Pi 4B, the system is fully solar-compatible and satisfies SCADA deadlines. 3. Usability: Human-readable threat alerts and interactive CLI wizards eliminate operational friction for non-specialist mine technicians. 4. Reliability: 75/75 automated unit tests validate mathematical stability across continuous operational cycles. 5. Security: The edge-native architecture operates completely offline without exposing telemetry to third-party cloud vulnerabilities.

CHAPTER 5: DISCUSSION AND CRITICAL ANALYSIS

5.1 Interpretation of Core Empirical Findings

All seven Design Objectives (DO1-DO7) established in Section 1.4 are formally validated by the experimental results. DO1 (< 100 ms latency) is exceeded by a wide margin: 0.76 ms mean execution is 131x below the SCADA ceiling and delivers a 207x speedup over the 157.66 ms Keras Float32 baseline. DO2 (< 1.0 MB model size) is satisfied at 0.82 MB. DO3 (>= 65% accuracy) is achieved at 70.56%. DO4 (>= 70% feature pruning) is achieved at 75.61%. DO5 (>= 75% accuracy floor) is achieved at 92.31% RF cross-validation accuracy. DO6 (offline edge autonomy) is validated via self-contained TFLite execution. DO7 (rapid deployment) is validated with a mean setup time of 2m 14s in user testing.

The 207x latency speedup is the single most transformative operational finding. At 0.76 ms, the system evaluates over 1,300 network flow samples per second on a single ARM core, enabling continuous, non-blocking monitoring of high-throughput industrial subnets governing hundreds of PLCs and telemetry nodes. The full-feature baseline requiring 157.66 ms processes fewer than 7 samples per second, creating severe buffer bloat that violates safety-critical control deadlines.

The BWOA feature selection results demonstrate clear semantic coherence: `src_bytes` (Gini=0.2451) and `error_rate` (0.1185) capture volumetric DoS attacks; `service` (0.1982) and `flag` (0.1420) distinguish industrial protocols (Modbus, DNP3) and TCP connection states; `same_srv_rate` and `diff_srv_rate` detect port scanning and lateral movement; and `hot` and `su_attempted` isolate unauthorized administrative privilege escalation. This confirms that BWOA selected physically meaningful features rather than random mathematical artifacts.

5.2 Comprehensive Accuracy-Latency-Size Trade-Off Justification

The 7.14% reduction in overall accuracy (from 77.70% baseline to 70.56% optimized) represents the fundamental engineering trade-off of this work. In industrial mining cybersecurity, this trade-off is completely justified across five distinct dimensions:

Deployability: 1. **Deployability Primacy:** An unoptimized model requiring 157.66 ms is completely unusable in real-time SCADA environments, rendering its theoretical 77.7% accuracy worthless. A 70.56% accurate model operating in 0.76 ms delivers actionable, real-time protection.

Benign Precision: 2. **Benign Precision Preservation:** False alarms that interrupt mineral production cost \$50,000/hr. The optimized model preserves 96.89% precision on normal traffic (compared to 97.12% baseline - a negligible 0.23% difference), ensuring operational continuity.

DoS Priority: 3. **DoS Recall Priority:** DoS flooding represents the most catastrophic threat to mining PLCs. The model preserves 89.04% recall on DoS attacks, capturing nearly 9 out of 10 volumetric intrusions.

Imbalance Reality: 4. **Dataset Imbalance Context:** Accuracy degradation is concentrated in extreme minority classes (U2R and R2L) where NSL-KDD contains only 52 training samples against 13,449 normal samples (259:1 imbalance), representing a dataset limitation rather than architectural failure.

Economic Accessibility: 5. **Industrial Cost Competitiveness:** Commercial OT security appliances cost \$20,000-\$50,000. This open-source framework delivers comparable detection capabilities on hardware costing under \$50.

5.3 Comparison with State-of-the-Art and Pareto Optimality

In the context of the related literature (Table 2.1), this work establishes a new Pareto-optimal operating point for industrial edge intrusion detection. Monolithic deep learning models (Kim et al., 2016; Ahmad et al., 2021) achieve higher raw accuracy but require gigabyte-scale RAM and multi-hundred-millisecond execution times that are incompatible with low-cost edge gateways. Shallow ML approaches (Ghosh et al., 2022) achieve acceptable speed but are limited to binary classification on balanced datasets. This research delivers the first system combining 5-class discrimination, sub-millisecond latency, sub-megabyte model size, and a complete production software ecosystem on physical 1GB RAM hardware.

5.4 Alignment with UN Sustainable Development Goals (SDGs)

SDG 9: SDG 9 (Industry, Innovation, and Infrastructure): Delivers open-source, resilient cybersecurity infrastructure tailored for resource-constrained extractive industries in developing economies.

SDG 8: SDG 8 (Decent Work and Economic Growth): Protects underground miners against cyber-physical tampering with ventilation, dewatering, and toxic gas monitoring systems, preserving human life.

SDG 17: SDG 17 (Partnerships for the Goals): Exemplifies bilateral scientific collaboration between Ghanaian researchers (UEW) and Russian academic hosts (Saint Petersburg Mining University) under UNESCO auspices.

5.5 Threats to Validity

Internal: Internal Validity: BWOA convergence was verified across 5 independent random seeds, confirming consistent 10-feature selection with minimal accuracy variance (92.14% - 92.31%). No training data leaked into feature evaluation or test scoring.

External: External Validity: Initial evaluations relied on benchmark corpora (NSL-KDD, SWaT). Phase 1 field PCAP capture at partner concessions (Gold Fields Tarkwa) will further calibrate detection on proprietary Modbus traffic.

Construct: Construct Validity: Metrics were evaluated across all 22,544 held-out KDDTest+ samples using standard multi-class formulations, and qualitative findings were triangulated via expert review and structured UAT questionnaires.

Statistical: Statistical Conclusion Validity: All benchmarks report unweighted precision, recall, and Macro F1 on the full 22,544-sample test partition to prevent misleading optimism from majority class dominance.

CHAPTER 6: SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

6.1 Summary of the Study and Key Artifact Outcomes

This Design Science Research investigation addressed the critical cybersecurity vulnerability gap in digitalizing African and Russian mining operations. By combining a Binary Whale Optimization Algorithm (BWOA) with a hybrid spatial-temporal CNN-LSTM neural classifier and post-training Float16 quantization, we produced a highly optimized, edge-deployable intrusion detection artifact. The system prunes input dimensionality by 75.61% (10 features), achieves 70.56% multi-class accuracy on KDDTest+, 96.89% precision on benign traffic, 89.04% recall on DoS attacks, and executes single-sample inference in 0.76 milliseconds on a Raspberry Pi 4B edge node. This establishes a 207x latency speedup over baseline models, operating well within the strict sub-100ms control deadline of industrial SCADA systems.

6.2 Formal Answers to Research Questions

6.3 Practical Contributions of the Developed Artifact

- 1. Academic: Academic Contributions: Formulates the first systematic DSR framework integrating BWOA feature selection with constrained accuracy floors and quantized CNN-LSTM models for industrial subsoil cybersecurity.**
- 2. Industrial: Industrial Contributions: Delivers a production-ready, open-source intrusion detection system compatible with Raspberry Pi edge gateways and cloud SaaS dashboards, directly deployable across Gold Fields Tarkwa, AngloGold Ashanti, and Minerals Commission pilot sites.**
- 3. Social & Policy: Social & Policy Contributions: Directly advances UN Sustainable Development Goals (SDG 9, SDG 8, SDG 17), protecting miner lives from cyber-physical disasters and building local African engineering capacity.**

6.4 Actionable Recommendations

Short-Term: Short-Term (0-6 Months): Finalize Phase 1 live PCAP telemetry capture with partner mining concessions (Gold Fields Tarkwa, AngloGold Ashanti) to establish a domain-specific baseline dataset.

Medium-Term: Medium-Term (6-18 Months): Deploy pilot edge nodes across 3 mining facilities in Ghana, South Africa, and the DRC, integrating local alerts into existing plant distributed control systems (DCS).

Long-Term: Long-Term (18-36 Months): Establish a pan-African mining threat intelligence exchange and contribute open-source detection rules to the global industrial cybersecurity community.

6.5 Future Research Directions

- 1. Field Data: Phase 1 Field Dataset Collection: Complete on-site packet capture at African mining sites to create the first open-source, domain-specific Modbus/DNP3 mining intrusion benchmark dataset.**
- 2. Federated Learning: Federated Learning Integration: Implement decentralized federated learning across multiple mining concessions, enabling collaborative threat intelligence sharing without exposing proprietary operational telemetry.**
- 3. INT8 Quantization: INT8 Quantization for Microcontrollers: Investigate INT8 quantization with representative calibration to compress models below 0.5 MB for deployment on Cortex-M7 PLC microcontrollers.**
- 4. Adversarial Defense: Adversarial Hardening: Evaluate and harden the neural classifier against gradient-based adversarial packet perturbations designed to evade spatial-temporal filters.**
- 5. Explainable AI: Explainable AI (XAI) for Control Rooms: Integrate real-time SHAP feature attribution into the SaaS dashboard to provide operators with immediate, plain-language justifications for anomaly alerts.**
- 6. HIL Validation: Hardware-in-the-Loop SCADA Testbed: Validate physical actuator response times using simulated PLC testbeds running industrial water treatment and ventilation control loops.**
- 7. Graph Neural Networks: Graph Neural Network Architectures: Model mining SCADA network topologies as graphs to detect multi-hop lateral movement attacks invisible to single-flow classifiers.**
- 8. Compliance Logging: Blockchain Compliance Logging: Integrate immutable cryptographic audit trails to automate regulatory reporting for the Minerals Commission of Ghana and international ESG safety registries.**

6.6 Concluding Remarks

The digital transformation of the global mineral resources complex represents an unprecedented opportunity to drive operational efficiency, environmental sustainability, and worker safety. However, this potential can only be realized if critical cyber-physical infrastructures are safeguarded against malicious intrusion. This research provides a verified, accessible, and mathematically rigorous security foundation for the digital mines of tomorrow, bridging the gap between cutting-edge artificial intelligence and the real-world operational constraints of African and Russian subsoil operations.

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APPENDICES

APPENDIX A: Semi-Structured Practitioner Interview Guide

The following interview protocol was administered to operational technology engineers and security managers at Ghanaian mining operations during requirements gathering:

1. Operational Architecture: 'What industrial communication protocols (Modbus, DNP3, OPC-UA, Ethernet/IP) are currently deployed in your milling and tailings control loops?' **2. Hardware Constraints:** 'What are the computing specifications and connectivity constraints of your substation telemetry gateways?' **3. Latency Deadlines:** 'What is the maximum tolerable security evaluation delay before a control loop safety margin is violated?' **4. Threat Landscape:** 'Have you observed unauthorized setpoint modifications, unauthenticated polling, or volumetric network floods in your OT environment?' **5. Alert Usability:** 'What information must an intrusion detection alert contain for control room technicians to execute effective mitigation?'

APPENDIX B: Questionnaire for User Acceptance Testing (UAT)

Domain specialists scored the platform on a 1 to 5 Likert scale across five evaluation dimensions: 1. Threat Alert Clarity: 'Are the displayed attack classifications and confidence scores clear and actionable during live operational events?' (1 = Very Cryptic, 5 = Very Clear) 2. Real-Time Responsiveness: 'Does the live telemetry stream update with sufficient rapidity to provide meaningful situational awareness without lagging?' (1 = Very Slow, 5 = Sub-Second) 3. CLI Deployment Ergonomics: 'Can the sniffer CLI agent be installed and bound to a network adapter in less than five minutes?' (1 = Very Difficult, 5 = Very Easy) 4. Risk Triage Trust: 'Does the confidence percentage assist in distinguishing high-severity volumetric intrusions from benign operational shifts?' (1 = Untrusted, 5 = Highly Trusted) 5. Concession Suitability: 'Would you recommend this solution for edge deployment on low-power, bandwidth-constrained African mining installations?' (1 = Unsuitable, 5 = Highly Recommended)

APPENDIX C: Binary Whale Optimization Algorithm (BWOA) Mathematical Pseudocode

APPENDIX D: CNN-LSTM Hyperparameters & Layer Tensor Shapes Specification

APPENDIX E: Complete 41-Feature Pruning & CICFlowMeter-to-SCADA Mapping Table

APPENDIX F: Automated Test Suite & Verification Matrix (75/75 Pass)

APPENDIX G: User Manual & Step-by-Step CLI Sniffer Operation Guide

To operate the @mhiskall282/unesco-mine-sec-cli agent on an edge gateway: 1. Registry Configuration: Configure npm to resolve the @mhiskall282 scope from GitHub Packages: npm config set @mhiskall282:registry <https://npm.pkg.github.com>

2. Direct Execution: Run the sniffer interactively via npx without local installation: npx @mhiskall282/unesco-mine-sec-cli

3. Interactive Setup Wizard: Select the monitoring network interface (e.g., eth0, wlan0) and specify the API analysis endpoint.

4. Headless Production Daemon: For automated background operation via systemd on industrial gateways: unesco-mine-sec-cli --url http://127.0.0.1:8001/api/analyze --interface eth0 --interval 100

APPENDIX H: Core Source Code Implementations & Open-Source Artifacts

The complete open-source codebase, training notebooks, and validation scripts are maintained at: <https://github.com/mhiskall282/Securing-the-Digital-Mine-UNESCO-Project>

Core Module Architecture: * Metaheuristic Optimizer: src/feature_selection/bwoa.py * Spatial-Temporal Classifier: src/models/cnn_lstm.py * Float16 Quantization Engine: src/benchmarks/edge_benchmark.py * Industrial Sniffer CLI: npm-packet-scanner/index.js * Inference Microservice: src/api_service.py * Cloud CI/CD Pipeline: .github/workflows/npm-publish.yml * SaaS Monitoring Console: dashboard/ (Laravel 12 Livewire)

APPENDIX I: Comprehensive 14-Criterion System Evaluation Scorecard

APPENDIX J: Complete Epoch-by-Epoch Neural Network Training History (Epochs 1-38)

CNN-LSTM v3 training progression on Google Colab T4 GPU (TensorFlow 2.15, batch_size=256, initial lr=1e-3):

APPENDIX K: REST API OpenAPI Contract & Payload Specification

APPENDIX L: Glossary of Technical Terms & Acronyms

APPENDIX M: Phase 1 Field Data Collection & Sensor Deployment Methodology

Phase 1 field data capture is scheduled for execution across partner concessions (Gold Fields Tarkwa, AngloGold Ashanti Obuasi, and UEW Innovation Hub research sites). The deployment architecture utilizes passive Ethernet network taps connected to SPAN mirror ports on industrial substation switches. Raspberry Pi 4B edge nodes running CICFlowMeter extract bi-directional flow features in real time, encrypting records via AES-256 before staging them for model retraining. Data collection protocols strictly adhere to industrial non-disclosure agreements, ensuring that proprietary production volume figures and plant setpoint registers are fully anonymized prior to open-source dataset publication.